If we set $\omega = \sqrt[n]{r}e^{\frac{i(\theta+2k\pi)}{n}}$, the formula for the nth roots of a complex number has a nice geometric interpretation, as shown in Figure. Note that because $|\omega| = \sqrt[n]{r}$ the n roots all have the same modulus $\sqrt[n]{r}$ they all lie on a circle of radius $\sqrt[n]{r}$ with centre at the origin. Furthermore, the n roots are equally spaced along the circle, because successive n roots have arguments that differ by $\frac{2\pi}{n}$.

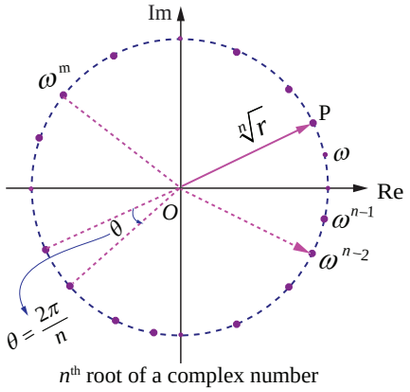


Fig. 2.44

Remark

(1) General form of de Moivre's Theorem

If x is rational, then $\cos x\theta + i\sin x\theta$ is one of the values of $(\cos\theta + i\sin\theta)^x$.

(2) Polar form of unit circle

$$\text{Let } z = e^{i\theta} = \cos\theta + i\sin\theta \text{. Then, we get}$$

$$|z|^2 = |\cos\theta + i\sin\theta|^2$$

$$\Rightarrow |x+iy|^2 = \cos^2\theta + \sin^2\theta = 1$$

$$\Rightarrow x^2 + y^2 = 1.$$

Therefore, $|z| = 1$ represents a unit circle (radius one) centre at the origin.

2.8.3 The nth roots of unity

The solutions of the equation $z^n = 1$, for positive values of integer n, are the n roots of the unity. In polar form the equation $z^n = 1$ can be written as

$$z^n = \cos(0 + 2k\pi) + i\sin(0 + 2k\pi) = e^{i2k\pi}, \quad k = 0, 1, 2, \ldots.$$

Using deMoivre's theorem, we find the nth roots of unity from the equation given below:

$$z = \left(\cos\left(\frac{2k\pi}{n}\right) + i\sin\left(\frac{2k\pi}{n}\right)\right) = e^{\frac{i2k\pi}{n}}, k = 0, 1, 2, 3, \ldots, n-1 \text{ .} \qquad \ldots (1)$$

Given a positive integer n, a complex number z is called an nth root of unity if and only if $z^n = 1$.

If we denote the complex number by ω, then

$$\omega = e^{\frac{2\pi i}{n}} = \cos\frac{2\pi i}{n} + i\sin\frac{2\pi i}{n}$$

$$\Rightarrow \omega^n = \left(e^{\frac{2\pi i}{n}}\right)^n = e^{2\pi i} = 1.$$

12th_Maths_EM_Vol1_CH 2_Complex Numbers.indd 87 16-12-2022 12:30:14